

Project Title
Smart Hospital Bed & Operation Room Scheduling System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Abstract

The **Hospital Bed and Operating Room (OR) Scheduling System** is an automated hospital management solution designed to improve the efficient utilization of hospital beds, operating rooms, and medical staff. The system assigns available beds to patients and schedules surgeries by considering bed availability, operating room availability, doctor schedules, surgery duration, and patient priority. Emergency cases are given higher priority to ensure timely treatment and reduce delays in critical situations.

The system replaces manual scheduling with an automated approach, reducing errors, double bookings, scheduling conflicts, and unnecessary patient waiting time. It also helps hospital administrators monitor and utilize available resources effectively. By coordinating patients, doctors, beds, and operating rooms through a centralized system, the hospital can improve workflow and make faster scheduling decisions.

Overall, the proposed system provides a reliable and efficient solution for hospital resource management, leading to better resource utilization, reduced waiting time, improved operational efficiency, and enhanced patient care.

Signature of the Guide